

Rational and Irrational Numbers on the Number Line

Worksheet 7A.02.05

Name _____

Date _____

Period _____

1. Sort the seven numbers into the table. Two of them carry a radical sign and are still rational.

0.125 $-\sqrt{12}$ -8 $-\sqrt{49}$ -2π $\sqrt[3]{-8}$ $-\frac{13}{3}$

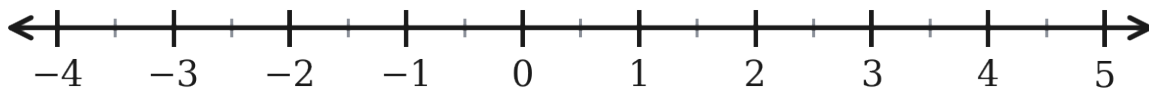
Rational	Irrational

For questions 2 through 7, compare each pair using $<$, $>$ or $=$.

2.	$\sqrt[3]{30}$		$\frac{7}{2}$
3.	$-2\sqrt{49}$		$-14.123\dots$
4.	$\pi^2 + 1$		3π
5.	$-\sqrt{64}$		$\sqrt[3]{-64}$
6.	$\sqrt{121}$		11
7.	$\frac{29}{6}$		$4.\overline{9}$

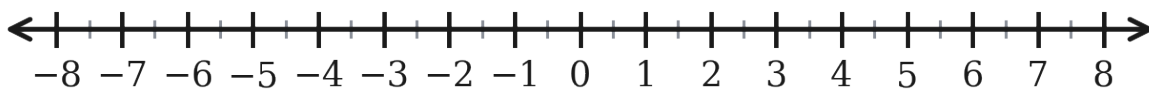
8. Order the numbers from least to greatest, then plot and label each one.

$$-3.\overline{25} \quad -\frac{7}{8} \quad \frac{\pi}{2} \quad 2\frac{1}{2} \quad \sqrt[3]{60} \quad \sqrt{22}$$



9. Order the numbers from least to greatest, then plot and label each one.

$$-\sqrt{54} \quad -2\pi \quad \sqrt[3]{-32} \quad \frac{19}{3} \quad 7.4$$



10. Which is greater, $-\sqrt{49}$ or -2π ? Explain how you decided, and say what a student who answers the other way is doing wrong.